Two solvers written from scratch, one trusted reference

Module 3 • vectorised Forward Euler and RK4 for $\dot{\mathbf{y}} = f(t, \mathbf{y})$, $\mathbf{y} = (s, e, i, r, d)$

Forward Euler one slope per step

$\mathbf{y}_{k+1} = \mathbf{y}_k + h\, f(t_k, \mathbf{y}_k)$; local error $O(h^2)$, global error $O(h)$

Classical Runge–Kutta (RK4) four slopes per step

$$k_1 = f(t_k, \mathbf{y}_k), \qquad k_2 = f\left(t_k + \tfrac{h}{2}, \mathbf{y}_k + \tfrac{h}{2} k_1\right),$$
$$k_3 = f\left(t_k + \tfrac{h}{2}, \mathbf{y}_k + \tfrac{h}{2} k_2\right), \quad k_4 = f(t_k + h, \mathbf{y}_k + h\, k_3)$$
$$\mathbf{y}_{k+1} = \mathbf{y}_k + \tfrac{h}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right)$$

- local error $O(h^5)$, global error $O(h^4)$
- both update all five compartments at once with NumPy arrays

Reference solution

- adaptive 8th-order Dormand–Prince (**DOP853**, SciPy)
- rtol 3×10^{-14}, atol 10^{-16}: close to the smallest SciPy accepts

Why so tight?

- a reference must be far more accurate than the solvers it judges
- with rtol $= 10^{-12}$, RK4's error at the smallest step reaches the reference's own ($\approx 10^{-12}$) and its measured order falls to **3.74**
- the fault is in the reference, not in RK4

Idea: Euler is the simplest method (order 1), RK4 the accurate workhorse (order 4). Both orders are known in advance, so the measured slopes test the code.

Two checks: conservation, and how accuracy is measured

Conservation of the total

$$\frac{d}{dt}(s + e + i + r + d) = 0$$

- the components of f sum to zero, so every Euler or RK4 step keeps $s + e + i + r + d = 1$, up to rounding

Result $t \in [0, 60]$, $h \in \{1,\ 0.25,\ 0.03125\}$

$$\max_k \left| \sum_j y_{k,j} - 1 \right| \ \leq\ 1.2 \times 10^{-15}$$

- a few multiples of machine epsilon (2.2×10^{-16})
- shows f is coded consistently; it does **not** measure accuracy

Experiment E1

- $\mathbf{y}_0 = (0.99,\ 0.005,\ 0.005,\ 0,\ 0)$ over $t \in [0, 60]$ days
- $h = 1,\ 0.5,\ 0.25,\ 0.125,\ 0.0625,\ 0.03125$ days

Global error: largest miss over all times and all five compartments

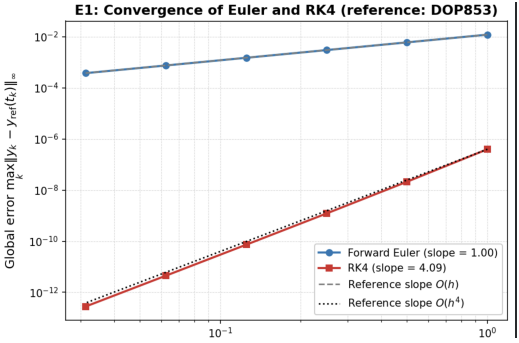
$$E(h) = \max_k \max_j \left| y_{k,j} - y_{\mathrm{ref},j}(t_k) \right|$$

Observed order: if $E(h) \approx Ch^p$, halving h gives $p \approx \log_2 \left[E(h)/E(h/2) \right]$; a least-squares fit of $\log E$ on $\log h$ gives one overall slope.

Two separate questions: conservation tests that the equations are coded consistently; convergence tests that the error shrinks at the theoretical rate.

E1 result: Euler converges at order 1, RK4 at order 4

Module 3 • global error against the DOP853 reference, log–log axes



reference lines: slopes 1 and 4

h (days)	Euler error	order	RK4 error	order
1	1.22×10^{-2}	1.004	4.00×10^{-7}	4.225
0.5	6.09×10^{-3}	1.002	2.14×10^{-8}	4.124
0.25	3.04×10^{-3}	1.001	1.23×10^{-9}	4.061
0.125	1.52×10^{-3}	1.001	7.34×10^{-11}	4.031
0.0625	7.59×10^{-4}	1.000	4.49×10^{-12}	4.015
0.03125	3.80×10^{-4}	–	2.78×10^{-13}	–
fitted slope		**1.001**		**4.085**

- Euler: error **halves** when h halves
- RK4: error falls by a factor of about **16**
- RK4's order sits slightly above 4 for large steps and approaches 4 as h shrinks

Result: observed orders 1.001 and 4.085 match the theory (1 and 4). At $h = 1$ day RK4 is about 3×10^4 times more accurate than Euler for four times the work per step, so RK4 is used for forward simulation.